

JUnit Basic Testing

Exercise 1: Setting Up JUnit

pom.xml Dependency :

```
<dependency>
  <groupId>junit</groupId>
  <artifactId>junit</artifactId>
  <version>4.13.2</version>
  <scope>test</scope>
</dependency>
```

Simple Java Class :

```
public class Calculator {
    public int add(int a, int b) {
        return a + b;
    }
}
```

JUnit Test Class :

```
import org.junit.Test;
import static org.junit.Assert.*;

public class CalculatorTest {

    @Test
    public void testAdd() {
        Calculator c = new Calculator();
        assertEquals(5, c.add(2, 3));
    }
}
```

Output :

Tests run: 1
Failures: 0

Exercise 2: Writing Basic JUnit Tests

Java Class :

```
public class Calculator {

    public int add(int a, int b) {
        return a + b;
    }

    public int multiply(int a, int b) {
        return a * b;
    }
}
```

```
}  
}
```

JUnit Test Class :

```
import org.junit.Test;  
import static org.junit.Assert.*;  
  
public class CalculatorTest {  
  
    @Test  
    public void testAdd() {  
        Calculator c = new Calculator();  
        assertEquals(5, c.add(2, 3));  
    }  
  
    @Test  
    public void testMultiply() {  
        Calculator c = new Calculator();  
        assertEquals(6, c.multiply(2, 3));  
    }  
}
```

Output :

Tests run: 2
Failures: 0

Exercise 3: Assertions in JUnit

```
import org.junit.Test;  
import static org.junit.Assert.*;  
  
public class AssertionsTest {  
  
    @Test  
    public void testAssertions() {  
  
        // Assert Equals  
        assertEquals(5, 2 + 3);  
  
        // Assert True  
        assertTrue(5 > 3);  
  
        // Assert False  
        assertFalse(5 < 3);  
  
        // Assert Null  
        assertNull(null);  
  
        // Assert Not Null  
        assertNotNull(new Object());  
    }  
}
```

```
}
```

Output :

Tests run: 1

Failures: 0

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

Calculator Class :

```
public class Calculator {  
    public int add(int a, int b) {  
        return a + b;  
    }  
}
```

Test Class :

```
import org.junit.*;  
import static org.junit.Assert.*;  
  
public class CalculatorTest {  
  
    Calculator calc;  
  
    @Before  
    public void setUp() {  
        calc = new Calculator();  
        System.out.println("Setup");  
    }  
  
    @After  
    public void tearDown() {  
        System.out.println("Teardown");  
    }  
  
    @Test  
    public void testAdd() {  
  
        // Arrange  
        int a = 2;  
        int b = 3;  
  
        // Act  
        int result = calc.add(a, b);  
  
        // Assert  
        assertEquals(5, result);  
    }  
}
```

Output :

Setup
Teardown

Tests run: 1
Failures: 0